Memo - Endpoint of line segments, given Midpoint

Pg 3 Nos 4 – 8

4. a) $M(1;3)$ $A(4;5)$ $B(x;y)$

$$M(1;3) = \left(\frac{4+x}{2}; \frac{5+y}{2}\right)$$

To find 'x':

$$1 = \frac{4+x}{2}$$

$$2 = 4+x$$

$$-2 = x$$

To find 'y'

$$3 = \frac{5+y}{2}$$

$$6 = 5+y$$

$$1 = y$$

b) $M(5;4)$ $A(3;2)$ $B(x;y)$

$$M(5;4) = \left(\frac{3+x}{2}; \frac{2+y}{2}\right)$$

$$5 = \frac{3+x}{2}$$

$$10 = 3+x$$

$$7 = x$$

$$4 = \frac{2+y}{2}$$

$$8 = 2+y$$

$$6 = y$$

c) $M(-\frac{1}{2}; 2)$ $B(-5;6)$ $T(x;y)$

$$M(-\tfrac{1}{2};2) = \left(\frac{-5+x}{2}; \frac{6+y}{2}\right)$$

$$-\frac{1}{2} = \frac{-5+x}{2}$$

$$-1 = -5+x$$

$$4 = x$$

$$2 = \frac{6+y}{2}$$

$$4 = 6+y$$

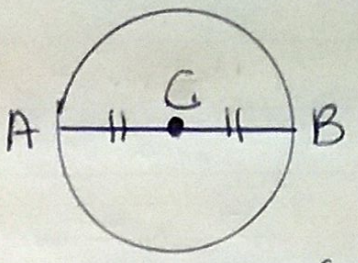
$$-2 = y$$

5. Centre $C(x:y)$ $A(-2:-1)$ $B(-1:9)$

$$C(x:y) = \left(\frac{-2-1}{2} ; \frac{-1+9}{2}\right)$$

$$\left(\frac{-3}{2} ; 4\right)$$

x y



$AM = MB$ (radii)

6

$M(-1:y) \rightarrow (0:-2)$ and $(x:8)$

$$M(-1:y) = \left(\frac{0+x}{2} ; \frac{-2+8}{2}\right)$$

$$(-1:y) = \left(\frac{x}{2} ; 3\right)$$

$-1 = \frac{x}{2}$ $\qquad$ $y = 3$

$-2 = x$

---

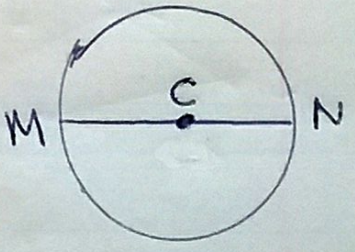
7. Centre $C(x:3)$ $M(5:-2)$ $N(-7:y)$

$$C(x:3) = \left(\frac{5-7}{2} ; \frac{-2+y}{2}\right)$$

$x = -1$ $\qquad$ $3 = \frac{-2+y}{2}$

$6 = -2+y$

$8 = y$



---

8 a) $x = -7$ and $y = -3$

b) $x = -7$ and $y = 8$

c) $x = -10$ and $y = 7$